

Real-Time Segmentation for Autonomous Robotic Manipulation

Vinith Reddy
Bennet University

1. Introduction

The intersection of computer vision and robotic surgery has become an important research area that makes surgical procedures safer, more autonomous, and more accurate. This literature survey discusses the main studies that support the paper "Real-Time Video Segmentation for Autonomous Robotic Manipulation, " which introduces a lightweight semantic segmentation method for the da Vinci Surgical System. The survey is divided into three main themes: (1) foundation models for segmentation, (2) robotic manipulation scene segmentation, and (3) real-time semantic segmentation architectures. These themes together shed light on the technical difficulties and design decisions that lead to the paper's two-stage pipeline employing SAM2 for pseudo-label generation and a small U-Net for real-time use.

2. Foundation Models for Segmentation

2.1 Segment Anything Model (SAM) and SAM2

Meta AI's Segment Anything Model (SAM) introduces a totally new way of doing image segmentation; it can perform any kind of segmentation from prompts over a wide range of visual domains. It has been trained on a gigantic dataset containing over one billion masks. SAM, by using points, boxes, and texts for prompting, generalizes very well even in zero-shot scenarios. Its newer version SAM2 brings this feature also to the video area by using a cross-frame memory attention module, which allows mask propagation between frames to be temporally consistent (Ravi et al. 2024).

This is very relevant for surgical video where for example, robotic arms and instruments can not only move quite a lot but also become partially occluded between frames. On the other hand, with about 600 ms per frame (around 1.6 FPS), SAM2's inference speed being so slow means it cannot simply be used for real-time robotic control that normally requires processing speeds of 30 Hz or more. Moreover, SAM2 necessitates a human-in-the-loop step of manually prompting for each video. It is these two weaknesses of the solution that have led the paper's authors to the decision of using SAM2 only in an offline pseudo-labeling step and not as a live inference model.

2.2 MedSAM and MedSAM2: Domain Adaptation for Medical Imaging

Even though SAM and SAM2 have a very good ability to generalize, surgical scenes present quite a few domain-specific challenges such as specular reflections, instruments obscured by blood, low-contrast tissue boundaries, and less-differentiated instruments from the background. Therefore, MedSAM and its successor that is also capable of handling video MedSAM2 adjust SAM2's architecture just on the medical and surgical imaging datasets, thus obtaining better handling of domain-specific alterations and structures (Zhu et al. 2024).

On the other hand, MedSAM2 still has the computational heaviness of the original SAM2 backbone, so it is as equally incapable of being deployed in real-time as before. These models are still research-grade tools that need GPU acceleration and cannot meet the sub-33 ms per frame latency budget necessary for closed-loop robotic control. The authors of this paper identify MedSAM2 as a related but hardly feasible alternative, thus their decision to use its supervision as a basis for the lightweight U-Net was further justified.

2.3 CLIPSeg: Zero-Shot Segmentation via Language-Image Prompts

CLIPSeg (Luddecke and Ecker, 2022) introduces a way to segment images without any prior training of the model, simply based on prompts that are given in natural language. The method takes advantage of the shared embedding space of images and texts by CLIP to create segmentation masks without the need for any training of the model on the specific task. Such a degree of versatility comes in handy in cases when the availability of labeled data is a major bottleneck, since one only needs to put target classes into words (for example, "robotic arm" or "small cylindrical object").

In reality though the performance of CLIPSeg on surgical scenes is highly inferior to that of masks generated by SAM2. The article uses CLIPSeg as a baseline and shows a Dice score of 81.3% with an inference time of 150 ms per frame (6.6 FPS) both measurements are worse than the U-Net method's. The main drawback is the training dataset of CLIPSeg, which misses out on a lot of surgical and robotic terms. This makes prompt tuning not dependable and leads to the creation of imperfect masks that cannot be used as imitation learning policy inputs.

3. Robotic Manipulation Scene Segmentation

3.1 Surgical Tool Segmentation: The EndoVis Challenge

The 2018 Robotic Scene Segmentation Challenge (Allan et al. 2020), also known as the EndoVis benchmark, offers a unified platform for assessing surgical instrument and scene segmentation with the da Vinci Surgical System. This set features detailed labeled video frames in RGB format from various robotic surgeries, marking ground-truth classes for surgical tools like bipolar forceps, prograsp forceps, large needle drivers, and vessel sealers.

EndoVis has grown to be one of the most important surgical AI benchmarks, helping to measure different surgical anatomy segmentation method on the same levels. The paper that is being reviewed gets its part of idea from EndoVis, most notably the aspects of laparoscopic challenges like occlusions, tool-tissue confusions, motion blurs,

and lighting changes that EndoVis highlights. Even though their dataset is a collection from the Stanford Robotics Center that they compiled themselves and not from EndoVis, the complexity of the scene and segmentation difficulties they face are pretty much the same.

3.2 U-Net and Its Variants for Surgical Scene Segmentation

Based on the context, the author probably wants to explain some technical features of U-Net related to the usage of skip connections in surgical image segmentation. The following is a rewritten humanized version of the given text that conveys the same technical details. The U-Net model which was first designed to segment biomedical images, features a symmetric encoder-decoder architecture with skip connections. These skip connections concatenate the high-resolution feature maps from the encoder to the upsampled decoder outputs (Siddique et al. 2020). Such a structure perfectly fits the task of surgical segmentation. The skip connections help to keep the precise details of the spatial structure, including the thin instrument shafts or the small manipulated objects, which might be lost during multiple pooling operations.

Siddique et al. (2020) have carried out an extensive review of U-Net derivatives used in medical image segmentation. The major architectural modifications are U-Net++ that implements dense skip connections for multi-scale feature fusion, and Attention U-Net that incorporates spatial attention gates to highlight relevant features while suppressing background ones. In scenarios involving very few data which are typical in surgical robotics due to the high cost of annotation these variants have been reported to show better generalization while maintaining high speed. The paper under review adopts a minimalistic two-block U-Net compact model that is trained using pseudo-labels derived through SAM2. They achieve a Dice score as high as 95.6% when matching SAM2's output and the inference time on a GPU is 10 ms (equivalent to 100 FPS). The authors here intentionally steer clear of very deep or complicated U-Net variants so as to allow for deployment on the limited-resource robotic hardware, while at the same time, they are able to produce segmentation results good enough to be utilized for downstream imitation learning.

3.3 Real-Time Instrument Tracking in Robot-Assisted Surgery

Qiu et al. (2019) put forward a multi-domain convolutional neural network for real-time surgical instrument tracking in robot-assisted surgery. Their method targets the difficulty of reliably localizing instruments even in very difficult conditions like partial occlusion, motion blur, and specular highlights. Besides the fact their CNN-based perception modules can be run in near-real-time when thoroughly optimized, their work also directly serves as a reference for the review paper's goal of developing a segmentation module that runs at 30 Hz. One major design issue pointed out by Qiu et al. is the necessity of linking segmentation outputs with downstream control policies, which means that segmentation outputs must be not only spatially accurate but also temporally consistent across frames. The reviewed paper also reflects this understanding in its choice of training on SAM2-generated pseudo-labels propagated through video by cross-frame memory attention, which guarantees the production of consistent quality masks for all frames in a demonstration.

4. Real-Time Semantic Segmentation Architectures

4.1 ENet: Early Work in Efficient Deep Segmentation

ENet (Paszke et al. 2016) put forward one of the earliest deep learning architectures that are end-to-end and explicitly made for real-time semantic segmentation. The main idea was that by substantially decreasing the size of the model with early downsampling, asymmetric convolutions, and a very distinct pruning of the decoder, ENet was able to achieve a level of inference speed that is suitable for embedded deployment while also having high accuracy on the usually used benchmarks like CamVid and Cityscapes.

The main contribution of ENet was to show segmentation performance could remain high while also achieving fast speed. A well thought out network architecture however can give you both results. This idea became a cornerstone of the paper under consideration. It argued that a simple U-Net being trained under top-notch guidance can come very close in performance to the large foundation model in the field of segmentation quality, while still being very cheap computationally at the time of being deployed.

4.2 BiSeNet: Bilateral Segmentation for High-Resolution Real-Time Inference

BiSeNet (Yu et al. 2018) is based on ENet with the idea of splitting the process of preserving spatial details and extracting contextual features into two distinct branches: one is the Spatial Path that keeps high-resolution feature maps, and the other is the Context Path which employs fast downsampling for extracting global context in a highly efficient manner. Subsequently, a Feature Fusion Module is used to combine the results of these two branches, which allows for high-resolution segmentation with low latency.

BiSeNet achieves state-of-the-art performance among real-time segmentation methods on CityScapes, showing that the spatial-contextual trade-off can be fixed architecturally instead of by brute-force parameter reduction. Although the paper under review is using a standard U-Net rather than a BiSeNet-style dual-branch design, the general idea that real-time segmentation needs intentional architectural decisions which balance resolution, context, and speed is the main point of both works.

4.3 Fast-SCNN: Fast Semantic Segmentation for Mobile and Embedded Devices

By using a two-branch encoder, Fast-SCNN (Poudel et al. 2019) focuses on mobile and embedded inference. A shallow learning-to-downsample module is the first branch that is used for fast spatial reduction, while a deep feature extractor is the second branch that is used for semantic context. Also, a lightweight decoder combines multi-scale features and generates high-quality segmentation masks with a very low number of parameters. Fast-SCNN is a very good example when one talks about deploying robots in scenarios with low compute resources, e.g. edge processing of surgical robots without dedicated GPUs. The paper under review shows that the U-Net can have 80 ms

inference on CPU (12.5 FPS) and 10 ms on GPU (100 FPS) thus in the robotics deployment context, comparing favorably to Fast-SCNN-class models. Future extensions of the reviewed work may consider Fast-SCNN-style architectures if deployment on embedded hardware without GPU acceleration is required.

5. Synthesis and Gap Analysis

The paper under review is obviously very well-grounded in the context described above. SAM2 is a foundation model that gives a level of mask quality that has never been reached before. However, the model is so resource-intensive that running it in real time is out of question. MedSAM2, which is a domain-specific model, is highly tuned for surgical applications but it is not a solution for reducing the delay. CLIPSeg, a text-prompted model, is great for convenience but it is less accurate when it comes to rare or less frequent domains. Speed is the advantage of traditional CV algorithms but, unlike the other methods, they cannot handle the diversity of surgical scenes.

Lightweight real-time architectures including ENet BiSeNet Fast-SCNN plus some other demonstrate that speed and accuracy are not necessarily opposite sides of a coin, yet they are normally in need of a large data corpus with annotations to training efficacy level - this is a very big barrier in surgical robotics. The paper under review makes use of SAM2 as an automated annotation machine and thus gets rid of the manual labeling need, on top of that provides the high quality supervision signals that are required for training the compact U-Net quite effectively.

The resulting pipeline of combining SAM2 for generating offline pseudo-labels and U-Net for real-time inference is a well-thought out combination of complementary strengths from different parts of the literature. This method aligns with a general direction in machine learning of knowledge distillation and teacher-student learning. In these setups, a powerful but slow teacher model(SAM2)supervises a compact, fast student model(U-Net)that can be used in latency-critical environments.

The main problem brought to light by this research is that there is no labeled ground-truth dataset available for the exact tasks and hardware that were used. The only way it is partially solved is by pseudo-labeling. Further research however should think of adding actual human annotations on a portion of the data in a way that can be used to check the quality of pseudo-labels, especially at occlusion boundaries and in highly blurred frames due to motion.

6. References

-
1. Allan, M., Kondo, S., Bodenstedt, S., Leger, S., et al. (2020). 2018 Robotic Scene Segmentation Challenge. arXiv preprint arXiv:2001.11190.
 2. Biewald, L. (2020). Experiment Tracking with Weights and Biases. <https://www.wandb.com/>
 3. Luddecke, T. and Ecker, A. (2022). Image Segmentation Using Text and Image Prompts. Proceedings of CVPR, pages 7086–7096.

4. Paszke, A., Chaurasia, A., Kim, S., and Culurciello, E. (2016). ENet: A Deep Neural Network Architecture for Real-Time Semantic Segmentation. arXiv preprint arXiv:1606.02147.
5. Poudel, R. P., Liwicki, S., and Cipolla, R. (2019). Fast-SCNN: Fast Semantic Segmentation Network. arXiv preprint arXiv:1902.04502.
6. Qiu, L., Li, C., and Ren, H. (2019). Real-Time Surgical Instrument Tracking in Robot-Assisted Surgery Using Multi-Domain Convolutional Neural Network. Healthcare Technology Letters, 6(6), 159–164.
7. Ravi, N., Gabeur, V., Hu, Y.-T., Hu, R., Ryali, C., Ma, T., ... and Feichtenhofer, C. (2024). SAM 2: Segment Anything in Images and Videos. arXiv preprint arXiv:2408.00714.
8. Siddique, N., Sidike, P., Elkin, C., and Devabhaktuni, V. (2020). U-Net and Its Variants for Medical Image Segmentation: Theory and Applications. arXiv preprint arXiv:2011.01118.
9. Yu, C., Wang, J., Peng, C., Gao, C., Yu, G., and Sang, N. (2018). BiSeNet: Bilateral Segmentation Network for Real-Time Semantic Segmentation. Proceedings of ECCV, pages 325–341.
10. Zhu, J., Hamdi, A., Qi, Y., Jin, Y., and Wu, J. (2024). Medical SAM 2: Segment Medical Images as Video via Segment Anything Model 2. arXiv preprint arXiv:2408.00874.